

HAND MUSCLE

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I. INTRODUCTION:

The muscles of the hand occupy the palmar region and are divided into three compartments:

- Lateral compartment: contains the muscles of the thenar eminence.
- Medial compartment: contains the muscles of the hypothenar eminence.
- Intermediate compartment.

II. DESCRIPTIVE ANATOMY

A. LATERAL COMPARTMENT: THENAR EMINENCE

Includes muscles involved in thumb movement.

1. Short Abductor of the Thumb

- Origin: Tubercle of the scaphoid and anterior surface of the flexor retinaculum.
- Belly: Fusiform muscle extending from the carpal region to the thumb.
- Insertion: Base of the proximal phalanx of the thumb.
- Action: Abduction of the thumb.
- Innervation: Median nerve.

2. Opponens Pollicis

- Origin: Tubercle of the trapezium and flexor retinaculum.
- Belly: Short, triangular muscle extending from the carpal region to the 1st metacarpal.
- Insertion: Lateral border of the 1st metacarpal.
- Action: Opposition of the thumb.
- Innervation: Median nerve.

3. Short Flexor of the Thumb

- Origin: Superficial head: Tubercle of the trapezium and flexor retinaculum. Deep head: Trapezoid and capitate.

- Belly: Triangular muscle extending from the carpal region to the thumb.
- Insertion: Base of the proximal phalanx of the thumb.
- Action: Flexion of the proximal phalanx, flexion and adduction of the thumb.
- Innervation: Superficial head: Median nerve. Deep head: Ulnar nerve.

4. Adductor Pollicis

- Origin: Oblique head: Trapezoid and capitate. Transverse head: 2nd and 3rd metacarpals.
- Belly: Triangular muscle with a medial base.
- Insertion: Base of the proximal phalanx of the thumb.
- Action: Adduction of the thumb.
- Innervation: Ulnar nerve.

B. MEDIAL COMPARTMENT: HYPOTHENAR EMINENCE

Formed by muscles related to the little finger.

1. Palmaris Brevis

- Origin: Palmar aponeurosis.
- Belly: Quadrilateral muscle extending transversely from the deep skin to the palmar aponeurosis.
- Insertion: Skin of the hypthenar eminence.
- Action: Wrinkles the skin, hollowing the palm.
- Innervation: Ulnar nerve.

2. Abductor Digiti Minimi

- Origin: Pisiform and flexor retinaculum.
- Belly: Fusiform muscle extending from the carpal region to the little finger.
- Insertion: Base of the proximal phalanx of the little finger.
- Action: Abduction of the little finger.
- Innervation: Ulnar nerve.

3. Short Flexor of the Little Finger

- Origin: Hook of the hamate and flexor retinaculum.
- Belly: Small fusiform muscle.
- Insertion: Base of the proximal phalanx of the little finger.
- Action: Flexion of the proximal phalanx and the little finger.
- Innervation: Ulnar nerve.

4. Opponens Digiti Minimi

- Origin: Hook of the hamate and flexor retinaculum.
- Belly: Small triangular muscle.
- Insertion: Medial border of the 5th metacarpal.
- Action: Opposition of the little finger with the thumb.
- Innervation: Ulnar nerve.

C. INTERMEDIATE (MIDDLE) COMPARTMENT:

Consists of two layers.

a) Anterior Layer:

1. Lumbrical Muscles (4)

- Origin: Tendons of the flexor digitorum profundus.
- Belly: Four small fusiform muscles.
- Insertion: Tendons of the extensor digitorum.
- Action: Flexion of the proximal phalanx, extension of the middle and distal phalanges.
- Innervation: 1st and 2nd: Median nerve; 3rd and 4th: Ulnar nerve.

b) Posterior Layer:

2. Palmar Interossei Muscles

- Origin: Axial surfaces of the 1st, 2nd, 4th, and 5th metacarpals.
- Belly: Small fusiform muscles extending from the metacarpal to the phalangeal region.
- Insertion: Bases of the proximal phalanges and the corresponding extensor tendons.
- Action: Same as the lumbricals + adduction of the fingers.
- Innervation: Ulnar nerve.

3. Dorsal Interossei Muscles

- Origin: Interosseous surfaces of the metacarpals (double origin).
- Belly: Small fusiform muscles.
- Insertion: Bases of the proximal phalanges of the 2nd, 3rd, and 4th fingers and corresponding extensor tendons.
- Action: Same as the lumbricals + abduction of the fingers.
- Innervation: Ulnar nerve.

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